

SED

Student Experiment Documentation

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Team Name: CLOUDS

Experiment Title: **Calcium Carbonate Light Observation Under Stratospheric Dynamics and Scattering**



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PREFACE

The REXUS/BEXUS program is realized under a bilateral Agency Agreement between the German Aerospace Center (DLR) and the Swedish National Space Agency (SNSA). The Swedish share of the payload has been made available to students from other European countries through a collaboration with the European Space Agency (ESA). EuroLaunch, a cooperation between the Swedish Space Corporation (SSC) and the Mobile Rocket Base (MORABA) of DLR, is responsible for the campaign management and operations of the launch vehicles. Experts from DLR, SSC, ZARM and ESA provide technical support to the student teams throughout the project. REXUS and BEXUS are launched from SSC, Esrange Space Center in northern Sweden.

CLOUDS (Calcium carbonate Light Observation Under stratospheric Dynamics and Scattering) is an experiment developed by a team of students from the Justus Liebig Universität Giessen in Germany. It aims to investigate the effects of calcium carbonate particles on solar radiation in the stratosphere, with implications for understanding and advancing Solar Radiation Management (SRM) techniques. CLOUDS is scheduled to fly aboard BEXUS 38 as part of cycle 17 of the Swedish-German program REXUS/BEXUS.

This document contains all relevant available information regarding CLOUDS, including the experiment's objectives and requirements, a breakdown of the experimental module's design, design verification procedures, a timeline, and plans for pre-launch campaign activities. Technical documents that pertain to the design are also included for reference as appendix sections.

Kommentiert [MN1]: updated

ABSTRACT

The CLOUDS experiment (Calcium Carbonate Light Observation Under Stratospheric Dynamics and Scattering) investigates the influence of calcium carbonate (CaCO_3) aerosols on incoming solar radiation in the stratosphere. The objective is to analyze, under realistic stratospheric conditions, how CaCO_3 particles modify the incident solar spectrum through scattering processes and to assess their potential applicability within the framework of Solar Radiation Management (SRM).

Solar Radiation Management refers to a class of climate intervention strategies aiming to reduce global temperatures by increasing the reflection of incoming solar radiation back into space. One of the most widely discussed approaches is Stratospheric Aerosol Injection (SAI), in which reflective particles are introduced into the stratosphere to enhance scattering of shortwave solar radiation and thereby reduce the Earth's radiative forcing.

The CLOUDS experiment focuses on spectral measurements in the 350–850 nm wavelength range. Conducted aboard a BEXUS stratospheric balloon at altitudes of approximately 25 km to 30 km, the experiment performs in-situ relative optical measurements using two parallel optical paths. One path serves as a particle-free reference, while the second passes through a sealed chamber containing dispersed CaCO_3 aerosols. Spectrometers record changes in spectral intensity before, during, and after particle release.

The expected results include measurable, potentially wavelength-dependent variations in light intensity attributable to scattering by CaCO_3 aerosols. The acquired data will enable a quantitative assessment of the particles' optical properties, thereby contributing to the evaluation of calcium carbonate as a reflective aerosol candidate. Furthermore, the experiment establishes a robust methodology for future stratospheric balloon-based studies of aerosol–radiation interactions relevant to SRM research.

1 INTRODUCTION

1.1 Scientific/Technical Background

The study of atmospheric aerosols and their impact on solar radiation is central to understanding climate dynamics and evaluating novel climate intervention strategies. Among these, Solar Radiation Management (SRM) refers to a set of approaches designed to reduce the amount of incoming solar energy absorbed by the Earth system, thereby partially offsetting the warming effect of anthropogenic greenhouse gas emissions. Unlike mitigation strategies that reduce CO₂ concentrations, SRM does not address the root cause of climate change but aims to alter the planet's radiative balance by increasing reflectivity or scattering of sunlight before it reaches the surface. SRM methods seek to rapidly influence global or regional temperatures, potentially buying time for deep decarbonization and carbon removal efforts to scale up. (<https://www.climate.gov/news-features/understanding-climate/solar-radiation-modification-noaa-state-science-factsheet>)

Among the proposed SRM techniques, Stratospheric Aerosol Injection (SAI) has received the most scientific attention and modeling effort. In SAI, reflective particles are introduced into the stratosphere, typically at altitudes between ~20 km and 30 km, where they can remain for months to years. These particles scatter a fraction of incoming solar radiation back into space, reducing net radiative forcing and surface heating. This concept draws on natural analogs, most importantly the observed cooling following large volcanic eruptions. For example, the 1991 eruption of Mount Pinatubo injected vast quantities of sulfur dioxide into the stratosphere, forming sulfate aerosols that measurably reduced mean surface temperatures by up to 0.5°C for about a year. (<https://www.climate.gov/news-features/understanding-climate/solar-radiation-modification-noaa-state-science-factsheet>)

1.1.1 Challenges with Sulfate Aerosols

Historically, the most studied candidate for SAI has been sulfate aerosols formed from precursor gases like sulfur dioxide (SO₂). Sulfate's prominence arises partly because of its resemblance to volcanic aerosols and relatively well-understood radiative properties. However, multiple undesirable side effects have been identified in climate model simulations and atmospheric chemistry analyses:

- **Stratospheric heating:** Sulfate aerosols absorb infrared radiation, leading to warming of the lower stratosphere. This temperature change can alter circulation patterns and potentially disrupt weather systems.
- **Ozone depletion:** Sulfate particles provide surfaces for heterogeneous chemical reactions that accelerate the breakdown of stratospheric ozone, undermining the protective ozone layer. This has been observed after large sulfate injections in models and analog studies.
- **Diffuse radiation increase:** Sulfate clouds scatter sunlight broadly, increasing diffuse light at the surface, which can affect ecosystems and photosynthesis in ways that are not fully understood.

These drawbacks highlight just some of the risks and complexities inherent in deploying sulfate-based SAI. In addition, most of the research on sulfate aerosols relies on existing atmospheric chemistry knowledge and models tuned to volcanic eruptions, leaving gaps in understanding when conditions differ from natural events.

(<https://pubs.rsc.org/en/content/articlehtml/2024/ea/d3ea00134b>,
<https://pmc.ncbi.nlm.nih.gov/articles/PMC5206531/>,
<https://www.climate.gov/news-features/understanding-climate/solar-radiation-modification-noaa-state-science-factsheet>,
<https://agupubs.onlinelibrary.wiley.com/doi/epdf/10.1002/2016GL069258>)

1.1.2 Calcium Carbonate as a possible candidate

Because of the limitations associated with sulfate aerosols, researchers have explored alternative materials that might offer comparable radiative effects with fewer negative chemical and dynamic consequences. Among these candidates, calcium carbonate (CaCO_3) stands out for several theoretical advantages:

- Reduced stratospheric heating: Models that simulate the injection of solid particles like CaCO_3 (calcite) indicate lower warming of the stratosphere compared to equivalent cooling using sulfate aerosols. This decrease in heating arises from differences in infrared absorption properties between solid mineral particles and sulfuric acid droplets. In some simulations, solid particles such as calcite or alumina lead to significantly smaller changes in stratospheric temperature profiles. (<https://salatainstitute.harvard.edu/new-directions-in-solar-geoengineering-research/>, <https://www.nature.com/articles/s43247-025-02038-1>)
- Potential mitigation of ozone depletion: Because calcium carbonate surfaces interact differently with reactive stratospheric gases, theoretical work suggests that CaCO_3 could reduce or even counteract ozone loss compared to sulfate particles, which tend to catalyze ozone-destroying reactions. In modeling studies of solid aerosol injection, the changes in column ozone under calcite injection are often smaller and less negative than with sulfuric acid aerosols, though uncertainties remain significant. (<https://www.nature.com/articles/s43247-020-00058-7>)
- Lower diffuse radiation impacts: Some modeling studies report that solid particle injection may produce less diffuse radiation at the surface than sulfate aerosols, potentially reducing unintended impacts on ecosystems that depend on specific light regimes. (<https://salatainstitute.harvard.edu/new-directions-in-solar-geoengineering-research/>)

Despite these potential advantages, significant scientific uncertainties remain. Laboratory investigations of CaCO_3 under stratospheric conditions have shown that its chemical reactivity with ozone-depleting gases may differ from earlier assumptions, suggesting that existing models may overestimate or underestimate its effects on stratospheric chemistry. This underscores the need

for both controlled laboratory experiments and eventually field measurements to refine parameters used in climate and chemistry models. (<https://ui.adsabs.harvard.edu/abs/2017AGUFMGC43H1162K/abstract>)

1.1.3 Experimental Motivation and Hypothesis

We aim to directly build on this emerging scientific basis with our experiment: while modeling and laboratory studies have highlighted possible benefits of CaCO_3 and have constrained some reaction pathways, there is still a lack of direct observational data on how CaCO_3 particles interact with solar radiation under real stratospheric conditions. The central scientific question guiding this research is: “How does the presence of CaCO_3 particles alter the incoming solar spectrum in the stratosphere, and what does this imply for their potential use in solar radiation management?” This question addresses the need to understand the scattering behavior of CaCO_3 particles in situ, which is essential for assessing their suitability as a reflective aerosol in SRM strategies and for improving radiative forcing estimates in climate models.

To investigate this, several assumptions underpin the CLOUDS experiment. It is assumed that CaCO_3 particles remain stable during the measurement period, ensuring consistent scattering effects. Because CaCO_3 is expected to have low absorptive properties relative to its scattering efficiency, the optical impact is considered to be dominated by scattering, with absorption being negligible at the wavelengths of interest. The relatively low background aerosol levels in the stratosphere simplify interpretation, such that observed changes in spectral intensity can be attributed primarily to the released CaCO_3 particles rather than to natural variability. The measurement instruments are presumed to be linear, stable, and well-calibrated, enabling accurate detection of radiation changes across the targeted wavelength range.

Investigating only the optical wavelength region is a deliberate choice driven by both physical relevance and experimental feasibility. In the optical range (~350–850 nm) a large fraction of the incoming solar energy that reaches the stratosphere is concentrated, and this is precisely where particle scattering dominates over molecular absorption or emission, making it the most informative part of the spectrum for assessing how aerosols influence radiative transfer.

By extending beyond laboratory kinetics and modelling work toward direct optical measurements in the stratosphere, the CLOUDS experiment fills a crucial gap in the current research landscape and provides empirical data that can inform both theoretical assessments and model development for CaCO_3 -based SRM concepts.

1.2 Mission Statement

The CLOUDS experiment aims to address this knowledge gap by investigating the effect of calcium carbonate aerosols on solar radiation in the stratosphere. By measuring wavelength-dependent changes in light intensity in the range ~350–850 nm under natural sunlight during a BEXUS balloon flight, the

experiment provides realistic in-situ data that bridges the gap between laboratory studies and atmospheric conditions. These measurements will support the assessment of CaCO_3 as a potential SRM aerosol and form a scientific basis for defining experiment objectives and requirements for future geoengineering research.

1.3 Experiment Objectives

The primary objectives are as follows:

Primary scientific Objectives:

1. Measurement of Particle Effects on Solar Radiation: Achieve accurate measurements of wavelength-dependent changes in solar spectral intensity caused by calcium carbonate particles within the 350–850 nm range, providing direct in-situ observational data for aerosol radiative effects
2. Validation of SRM Potential: Assess the feasibility of using calcium carbonate particles as an effective reflective aerosol for Solar Radiation Management by relating measured optical effects to theoretical expectations and radiative forcing implications

Primary technical Objectives:

1. Successful Deployment: Ensure controlled release of calcium carbonate particles and reliable operation of the sealed chamber system, including controlled opening and closing of the chamber
2. Successful Operation: Maintain continuous particle dispersion and capture accurate spectral measurements throughout the mission

1.4 Experiment Concept

The CLOUDS experiment investigates the influence of calcium carbonate aerosol particles on solar radiation under stratospheric conditions using comparative optical measurements.

The core concept is based on a dual light path system, in which sunlight is measured along a particle-affected path and a reference path without particles. Key elements of the experiment include this dual light path system, a measuring chamber with a particle release, and dispersal mechanism, a set of environmental and motion sensors, and a data processing and storing unit.

During operation, CaCO_3 particles are released into the measuring chamber while the optical system records sunlight simultaneously in both light paths. Sensor data provides contextual information on environmental conditions and experiment orientation, and all measurements are collected and handled by the central data unit.

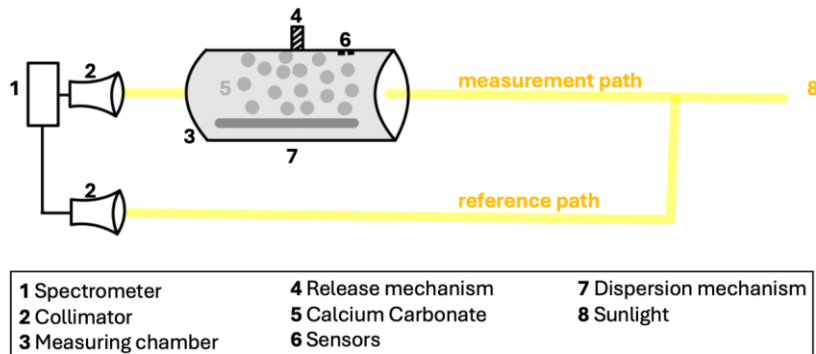


Figure 1-1 Fundamental experimental setup

1.5 Team Details

1.5.1 Contact Point

For any inquiries regarding the CLOUDS experiment, please contact our team with the following details:

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Team lead: Isabella Kaindl (mail: isabella.kaindl@physik.uni-giessen.de / i.kaindl@web.de, phone number: +49 177 6889 253)

Academic supervisor: Prof. Dr. Sangam Chatterjee (mail: sangam.chatterjee@physik.uni-giessen.de)

1.5.2 Team Members

Isabella Kaindl

Physics and Technology for Space Applications (bachelor, 3rd semester)

Team role: team leadership, overall concept development, integration of experiment subsystem

Expected workload: 10h/week

Academic credit: to be determined

David Kibler

Physics and Technology for Space Applications (bachelor, 3rd semester)

Team role: electronics and mechanical design, implementation of experimental setup

Expected workload: 10h/week

Academic credit: to be determined

Mathis Telschow

Data Science (bachelor, 5th semester)

Team role: software development, online presence, data acquisition and analysis

Expected workload: 10h/week

Academic credit: to be determined

Kai Wedekind

Physics and Technology for Space Applications (master, 1st semester)

Team role: optical system

Expected workload: 10h/week

Academic credit: to be determined

Nick Hofmann

Geography (bachelor, 5th semester)

Team role: geographical context, environmental background, data interpretation and evaluation

Expected workload: 1h/week

Academic credit: to be determined

Şafak Ürken

Teaching degree L3: Mathematics, Physics (state examination, 3rd semester)

Team role: didactic development, school outreach, external communication

Expected workload: 2h/week

Academic credit: to be determined

2 EXPERIMENT REQUIREMENTS AND CONSTRAINTS

2.1 Functional Requirements

Table 2-1: Functional Requirements

ID	Name	Requirement Description
F.1	Measurement of Solar Radiation intensity	The experiment shall measure the intensity of solar radiation in the stratosphere within the 350–850 nm range.
F.2	Spectral Distribution Measurement of Solar Radiation	The experiment shall measure the spectral distribution of solar radiation in the stratosphere within the 350–850 nm range.
F.3	Particle Dispersion	The experiment shall disperse CaCO_3 particles uniformly within the measurement chamber.
F.4	Particle Release	The experiment shall release the CaCO_3 in a controlled manner.
F.5	Measurement of Temperature	The experiment shall measure temperature during ascent.
F.6	Measurement of Humidity	The experiment shall measure relative humidity during ascent and inside the measurement chamber.
F7	Kamera	David

2.2 Performance Requirements

Table 2-2: Performance Requirements

ID	Name	Requirement Description
P.1	Intensity Measurement Accuracy	TBD, to be determined after selection of intensity measurement approach (e.g., photodiode vs. full spectrometer) David
P.2	Spectral Resolution / Discrete Spectral Sampling	TBD — depends on final optical sensor choice (full spectrum grating spectrometer vs. discrete wavelength filters/photodiodes) David

ID	Name	Requirement Description
P.3	Sampling Rate Spectrometer	The spectrometer acquires data at a sampling rate of at least 1 Hz.
P.4	Spatial Uniformity of Particle Distribution	TBD — target value to be set after dispersion mechanism prototyping and chamber flow characterization
P.5	Temporal Uniformity of Particle Distribution	TBD — target duration to be defined based on expected measurement window and settling/dispersion dynamics in chamber tests
P.6	Uniformity Achievement Time	The system shall achieve uniform dispersion within 5 minutes.
P.7	Uniformity Duration Particle Distribution	The particle distribution shall remain uniform for at least 3 minutes.
P.8	Measurement Range Temperature Sensor	The temperature measurement's range outside the balloon shall be between -60 °C and +30 °C.
P.9	Accuracy Temperature Sensor	The temperature measurements outside the balloon shall be made with an accuracy of ± 5 °C.
P.10	Sampling Rate Temperature Sensor	The temperature measurement outside the balloon shall be made at a rate of 1 measurement every second (1Hz)
P.11	Measurement Range Humidity Sensor	The humidity measurement range outside the balloon shall be between 0% and 100% relative humidity.
P.12	Accuracy Humidity Sensor	TBD, non-trivial for stratospheric conditions David
P.13	Sampling Rate Humidity Sensor	The humidity measurement outside the balloon shall be made at a rate of 1 measurement every second (1 Hz).

2.3 Design Requirements

Table 2-3: Design Requirements

ID	Name	Requirement Description
D.1	Temperature Profile	The experiment shall operate between -60°C and 30°C.

D.2	Vertical Shock Excitation	The system shall withstand a transient vertical shock excitation of -10 g .
D.3	Horizontal Shock Excitation	The system shall withstand a transient horizontal shock excitation of $\pm 5\text{ g}$.
D.4	Landing Velocity Shock	The system shall withstand a landing impact corresponding to a velocity of 7 m/s .
D.5	Transport Vibration Resistance	The system shall withstand unspecified random vibrations during transport phases.
D.6	Mounting in Independent Fixed Enclosures	The experiment shall be mountable in independent, fixed boxes which can be attached to the gondola.
D.7	E-Link Connector	The experiment power connector shall be an Amphenol RJF21B, Code A.
D.8	Power Connector	The experiment power connector shall be an Amphenol PT02E8-4P, Code A.
D.9	No Disturbance or Harm	The experiment shall not disturb or harm the launch vehicle.
D.10	Parallel Light Measurement	The experiment shall acquire data through parallel light measurements.
D.11	Light Intensity Equality Reference and Measurement Path	The light intensity in the reference path and the measurement path shall not differ by more than 10000% .
D.12	Coupling Efficiency per Phaser	The light coupling into each phaser (excluding the Axicon) shall result in a maximum of 20% loss per coupling.
D.13	Mass Limitation	The experiment shall not exceed a total mass of 10 kg .
D.14	Power Consumption Limit	The experiment shall not consume more than 150 Wh of energy during the mission.
D.15	Maximum Experiment Dimensions	The experiment shall have maximum dimensions of $60\text{ cm} \times 90\text{ cm} \times 40\text{ cm}$.

2.4 Operational Requirements

Table 2-4: Operational Requirements

ID	Name	Requirement Description
O.1	Modular Gondola Integration	The experiment shall be designed for modular removal or expansion, allowing flexible integration and maintenance.
O.2	Automatic Calcium Release / Default Experiment Execution	The experiment shall automatically release CaCO_3 particles twice in flight if the connection with the ground segment is lost or otherwise execute the full experiment sequence by default.
O.3	Onboard Data Storage	The experiment shall store all acquired data onboard the gondola
O.4	Controlled Downlink	The experiment downlink a predetermined amount of data to the ground segment.
O.5	Manual Pressure Equalization	The experiment shall provide a simple manual mechanism for pressure equalization.

2.5 Constraints

Table 2-5: Constraints

ID	Name	Requirement Description
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3 PROJECT PLANNING

3.1 Work Breakdown Structure (WBS)

To effectively plan and manage the experiment project, a preliminary Work Breakdown Structure (WBS) has been developed. The WBS provides a hierarchical framework that organizes the project into manageable work packages, ensuring that all technical, organizational, and administrative aspects are considered. This structure allows for clear assignment of responsibilities and helps track progress throughout the project lifecycle.

The following section presents a preliminary division of the project into work packages, which will serve as a basis for further planning, resource allocation, and detailed scheduling.



Figure 3-1: Work Breakdown Structure (WBS) for the CLOUDS experiment

For better readability, here it is again in enlarged form:

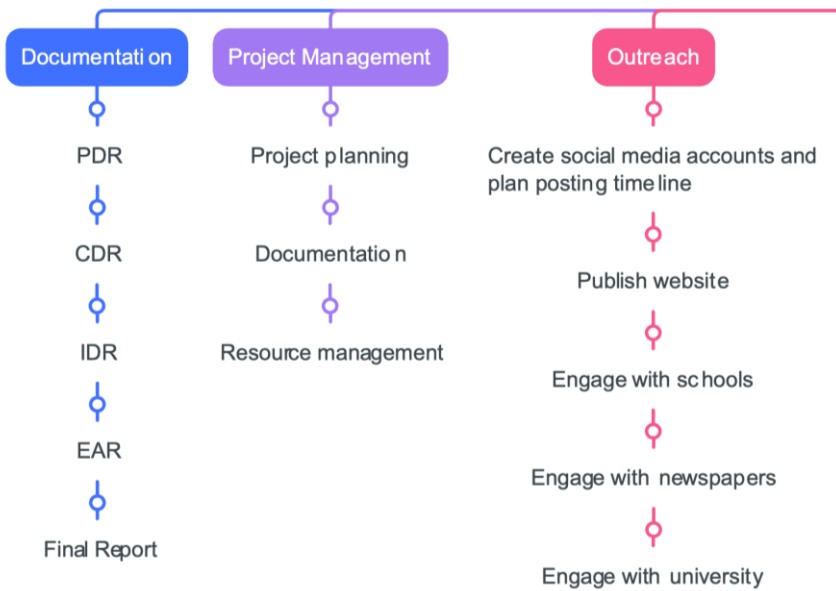


Figure 3-2: Documentation, Project Management, and Outreach within the WBS

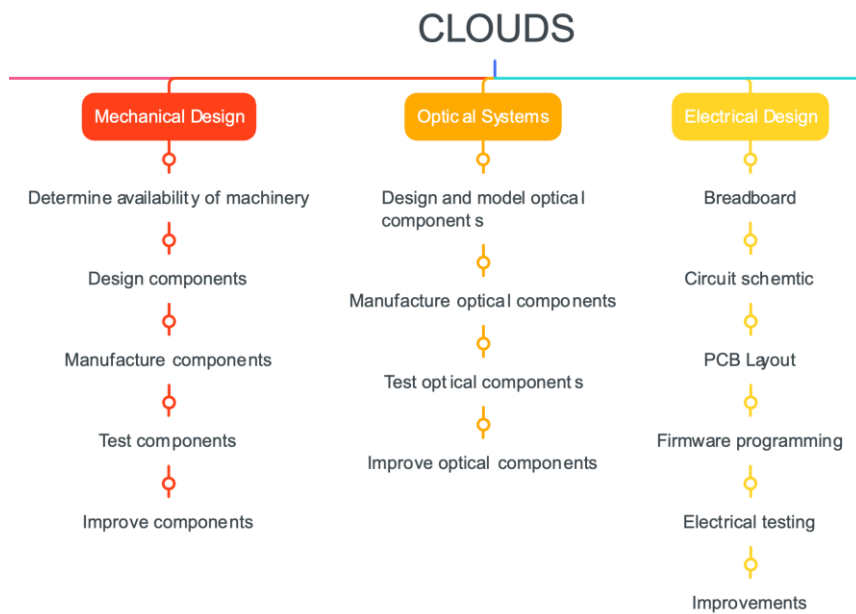


Figure 3-3: Mechanical Design, Optical Systems, and Electrical Design within the WBS



Figure 3-4: Software, Testing, and Data Analysis within the WBS

3.2 Schedule

A general Gantt chart presenting planned project schedule is shown below:

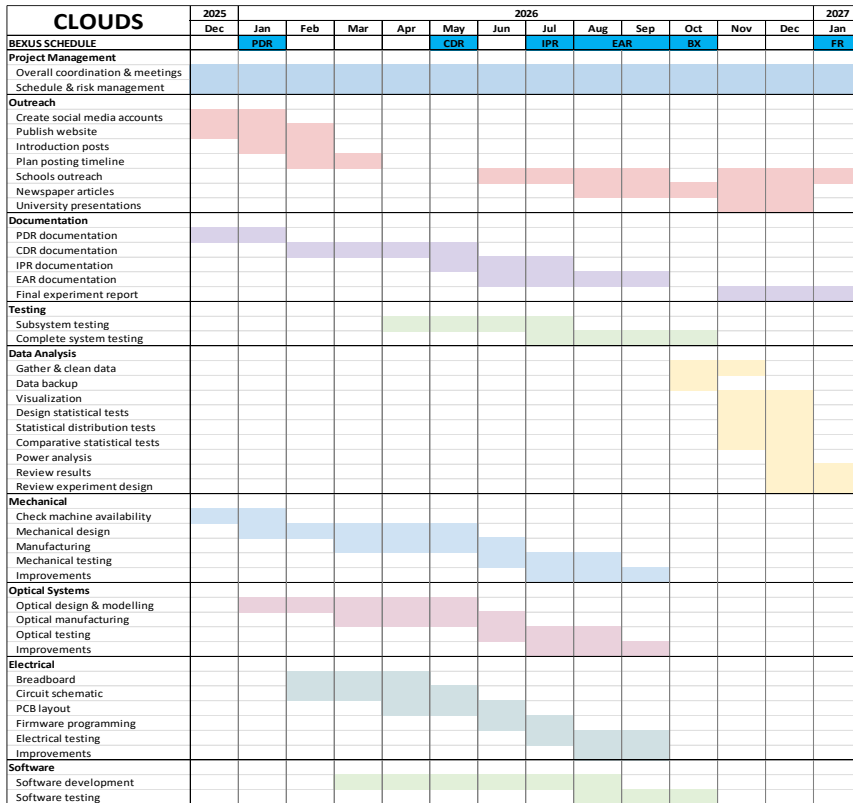


Figure 3-5: Gantt chart presenting planned project schedule

Already known campaign dates:

Table 3-1: Known campaign dates

26-30 Jan 2026	Preliminary Design Reviews (PDRs) & Students Training Week, SSC, Esrange Space Center
4-6 May 2026	BEXUS Critical Design Reviews (CDRs) and Soldering Course, ESA, ESTEC
Jul 2026	BEXUS Integration Progress Reviews (IPRs) at student team sites
Aug/Sep 2026	BEXUS Experiment Acceptance Reviews (EARs) at student team sites (ESA/SNSA) or ZARM-FAB Bremen (DLR)
2-12 Oct 2026	BEXUS Campaign (10 days), SSC, Esrange Space Center
Jan 2027	Delivery of the final BEXUS Experiments reports

3.3 Resources

3.3.1 Workforce

The team consists of 6 team members. The tables below present each team member with their relevant experience and skills.

Table 3-2: Team member Isabella Kaindl

Isabella Kaindl	
Role	Team leader, system integration
Experience	Second Board Member and member of the rocket body division at Spaceflight Rocketry Gießen (independent student rocket building group)
Declared availability	10h/week Not available: 20.02-28.02 exams, 01.03-07.03 vacation, 23.03-28.03 exams, 20.07.-05.08 exams
Replacement plan	Team lead: Mathis Telschow / Kai Wedekind, Engineering: David Kibler / Kai Wedekind

Table 3-3: Team member David Kibler

David Kibler	
Role	Electrical and Mechanical design
Declared availability	10h/week Not available: 20.02-28.02 exams, 01.03-07.03 vacation, 23.03-28.03 exams, 20.07.-05.08 exams
Replacement plan	Kai Wedekind

Table 3-4: Team member Mathis Telschow

Mathis Telschow	
Role	Data Analysis, Software
Experience	Web Developer at Birgit Schwind Creative GmbH and Softplan GmbH for 2 years Independent software projects for university purposes
Declared availability	10h/week Not available: 18.02.-28.02. exams, 15.06.-30.06. writing thesis, 20.07.-05.08. exams

Mathis Telschow	
Replacement plan	Data Analysis, Statistics: Recruit data science students at JLU Giessen Software: David Kibler

Table 3-5: Team member Kai Wedekind

Kai Wedekind	
Role	Mechanical and Optical design
Experience	7-month internship and bachelor thesis at ArianeGroup in Lampoldshausen, titled "Design of a Propellant Tank for a Water Propulsion System" Design of an EP thrust scale (facilitating optical measurements)
Declared availability	10h/week Not available: 16.03.-12.04., 21.09.-28.09.
Replacement plan	Mechanical engineering, manufacturing, and CAD: David Kibler, External communication and optical system design: Isabella Kaendl

Table 3-6: Team member Safak Ürken

Şafak Ürken	
Role	Outreach with focus on schools
Experience	Several years of teaching experience as a physics teacher
Declared availability	2h/week
Replacement plan	Mathis Telschow / Isabella Kaendl

Table 3-7: Team member Nick Hoffmann

Nick Hoffmann	
Role	Geographical context and environmental background.
Experience	Various projects as part of geography studies
Declared availability	1h/week
Replacement plan	Isabella Kaindl / Kai Wedekind

3.3.2 Budget

At the current stage of the project, several open points still affect the final budget planning of the experiment. As a result, the cost estimation presented here should be considered preliminary and subject to updates as the design and procurement processes progress.

A significant portion of the required optical components is expected to be provided by research groups at Justus Liebig University Giessen. Many optical elements can be borrowed or transferred from existing laboratory inventories, which substantially reduces procurement costs. However, due to the short timeframe between the end of the Christmas break and the submission deadline of the SED, several discussions with contributing research groups are still ongoing and could not yet be finalized.

Similarly, it is not yet fully clear which mechanical components can be manufactured within university workshops and to what extent raw materials or consumables can be sponsored. Clarification of these aspects is currently in progress through active communication with multiple university workshops and technical facilities.

The costs for the electronic components are currently estimated to amount to approximately 280€.

3.3.3 External Support

Isabella

Our experiment is supported by scientists from various disciplines who have already contributed their expertise during the preparation of this application. The project is accompanied by Prof. Dr. Sangam Chatterjee (Head of the Spectroscopy and Optics Group, 1st Institute of Physics, Justus Liebig University Giessen).

In addition, we are supported by the following experts:

Physics:

- Dr. Kristof Holste, Group Leader, Ion Thruster Research Group, 1st Institute of Physics (Justus Liebig University Giessen)
- Dr. Hans-Georg Zaunick, Senior Lecturer, Experimental Hadron Physics with a focus on Detector Physics (Justus Liebig University Giessen), who supported us through the design progress of our experiment.
- Dr. Roman Bergert, helped design an initial proof of concept experiment

Physics Education:

- Dr. Kathrin Steckenmesser-Sander, Senior Lecturer, Institute for Physics Education (Justus Liebig University Giessen)

Electronics:

- Prof. Dr. Jens Simon, Faculty of Electrical and Information Engineering (University of Applied Sciences Mittelhessen)
- Prof. Dr. Chris Volkmar, Faculty of Electrical and Information Engineering (University of Applied Sciences Mittelhessen)

Geography and Climate Science:

- Prof. Jürg Luterbacher, PhD, Chair in Climatology, Climate Dynamics and Climate Change, Institute of Geography (Justus Liebig University Giessen)

Statistics:

- Gerrit Eichner, PhD, senior academic advisor, Statistics advisor, Institute of Mathematics (Justus Liebig University Giessen)

Furthermore, we are supported by our university through the provision of an official email address and access to cloud storage services. In addition, we are in contact with Infineon and STMicroelectronics, both of which have expressed their readiness to support our project.

At present, we are continuing to strengthen our connections with the engineering department as well as with various workshops, with whom we are already in ongoing discussions.

EURECA

3.4 Outreach Approach

Mathis und Isabella

Table 3-8: Team social media links

Team	Instagram	LinkedIn	Website
CLOUDS	@clouds_bexus	clouds-bexus	cloudsbexus.de

Since climate change and potential geoengineering solutions will affect everyone, we are planning a multi-level outreach program to engage people of all ages.

Firstly, there are pupils. We already have connections with different schools in and around Giessen from previous programs. We want to build on this by talking about our project at a science cafe for students, where pupils can learn about all science- and technology-related topics once a week. Likewise, we will approach schools in Giessen to offer presentations to interested classes. As it is critical to us to present our findings and experimental results in an understandable way, one of our team members is a student teacher with a lot of teaching experience. He will ensure that our experimental content is processed in such a way that students can learn a lot from the presentations and retain our results in the long term.

By presenting our findings at university events, we can also raise awareness of BEXUS and our entire project at university level. We are already in contact with various departments eager to be kept up to date on our project.

Through an established connection with the local newspaper, Gießener Allgemeine Zeitung, we can share our experience with the public. We are planning to get in touch with other newspapers as well.

To keep everyone interested updated, we plan to create our own website and post our regular findings there. As it is important to include everyone interested in the project, we plan to use a mixture of scientific and popular scientific articles.

3.5 Risk Register

Isabella

This section identifies and assesses the risks associated with the CLOUDS experiment. Risks are systematically captured to ensure proper management through identification, evaluation, mitigation, and monitoring. The Risk Register will be updated in consecutive SED versions.

Risk categories, probabilities (P: A–E), and severities (S: 1–5) follow the guidelines:

Risk ID Categories:

- TC – Technical/Implementation
- MS – Mission (operational performance)
- SF – Safety
- VE – Vehicle
- PE – Personnel
- EN – Environmental
- SC – Scattering Measurement

Probability (P):

- A. Minimum – Almost impossible to occur
- B. Low – Small chance to occur
- C. Medium – Reasonable chance to occur
- D. High – Quite likely to occur
- E. Maximum – Certain to occur, maybe more than once

Severity (S):

- 1. Negligible – Minimal or no impact
- 2. Significant – Leads to reduced experiment performance
- 3. Major – Leads to failure of subsystem or loss of flight data
- 4. Critical – Leads to experiment failure or creates minor health hazards
- 5. Catastrophic – Leads to termination of the REXUS and/or BEXUS program, damage to the vehicle or injury to personnel

The rankings for probability (P) and severity (S) are combined to assess the overall risk classification, ranging from very low to very high and being colored green, yellow, orange or red according to the SED guidelines.

Overall Risk (P x S): Combined into classification (Minimum, Very Low, Low, Medium, High).

Table 3-9: Risk Register

ID	Risk & consequence	P	S	P x S	Prevention and/or Mitigation Action	New P	New S	New P x S
TC001	Particle clumping prevents uniform dispersion – leads to inaccurate scattering measurements.	C	3	Low	mindful handling; Dehumidification; ground tests.	B	2	Very Low
TC002	Particles unstable in stratosphere – alters optical properties.	C	2	Low	Literature Review; Pre-tests under simulated conditions; ground tests with particle sizes and membrane setup.	B	2	Very Low
TC003	Spectrometers non-linear or uncalibrated – incorrect intensity changes.	D	3	Medium	Documentation, Multiple calibrations	B	3	Low
TC004	Axicons or optics faulty – inaccurate light paths.	C	4	Medium	Prototype and ground tests; spare parts, shielding.	B	3	Low
TC005	Vibrating membrane failure – no particle dispersion.	C	4	Medium	Ground tests, cold tests, vibration tests.	B	4	Low
TC006	No particle release (calcium carbonate release failure).	B	4	Low	Valve tests; repeated actuation with manual signals.	A	4	Very Low
TC007	Sealing failure during float phase.	B	3	Low	Ground tests.	A	3	Very Low
TC008	Ventilation failure during descent or after landing.	B	1	Very Low	Valve tests.	A	1	Very Low
MS001	Data loss during float phase – incomplete spectra.	C	3	Low	On-board storage + downlink redundancy (200 kbps); automatic backup.	B	3	Low
MS002	IMU failure (failure to detect float phase).	C	4	Medium	Calibration and redundant IMU; software monitoring; experiment activation without IMU detection	A	2	Very Low
SF001	Handling of CaCO ₃ particles.	A	2	Very Low	Responsible handling procedures, calciumcarbonat is not harmful to humans	A	2	Very Low

ID	Risk & consequence	P	S	P x S	Prevention and/or Mitigation Action	New P	New S	New P x S
SF002	Magnetic field from coil – interference with other experiments.	B	4	Low	Shielding; tests for interference fields if necessary	A	2	Very Low

ID	Risk & consequence	P	S	P x S	Prevention and/or Mitigation Action	New P	New S	New P x S
VE001	Gondola contamination due to particle leakage.	C	4	Medium	Fail-safe valve; sealing tests.	A	2	Very Low
EN001	Aerosol release in stratosphere (minimal environmental impact).	B	1	Very low	Biodegradable CaCO ₃ ; small quantity (<10 g); no ejections planned.	A	1	Very Low
PE002	Team member dropout – project delays.	B	3	Medium	Redundant roles; knowledge transfer and documentation.	B	2	Very Low

4 EXPERIMENT DESCRIPTION

4.1 Experiment Setup

Isabella

The experimental setup consists of two parallel optical paths: a measurement path with calcium carbonate aerosols (particle path) and a reference path without particles. Both paths are designed to be geometrically and optically as identical as possible to minimize systematic effects and ensure direct comparability of the measurement data.

Solar light is first captured and collimated using specially designed axicons (conical lenses). These optics are dimensioned to account for the low solar elevation during the BEXUS flight and to provide stable light coupling independent of balloon rotation. After collimation, the light is split into two optical paths.

In the lower optical path, the light beam is directed horizontally into a measurement chamber where calcium carbonate particles are released. During ascent, the chamber remains open to the ambient environment, allowing real stratospheric pressure and temperature conditions to establish inside. Upon reaching the target altitude, the chamber is sealed, and the particles are uniformly distributed using a vibrating membrane. The light scattered by the aerosols then passes through optical filters and is coupled into a spectrometer detector that records the spectral intensity distribution.

The second optical path serves as the reference measurement. It is constructed analogously but contains no released particles. The light is initially guided through a fiber optic cable to the reference path, then passes through a chamber without aerosols, identical filters, and a similar detection unit. This setup enables simultaneous measurements of particle-influenced and particle-free solar light, allowing direct determination of relative changes in intensity and spectrum.

In addition to the optical system, the experiment is equipped with several sensor subsystems. Pressure and temperature sensors continuously monitor the environmental conditions inside the chamber as well as in the surrounding stratosphere. An Inertial Measurement Unit (IMU) provides data on the experiment's attitude, rotation, and acceleration. Particle sensors monitor the concentration and distribution of calcium carbonate aerosols.

Data acquisition and control of all subsystems are managed by a central electronics and computing unit. This unit coordinates the measurement sequences, controls particle release, and synchronizes sensor data with the optical measurements. The recorded data are stored onboard and partially transmitted to the ground segment via the BEXUS downlink interface during the flight.

4.2 Experiment Interfaces

4.2.1 Mechanical

Kai

The mechanical interface consists of two separate parts connected by two optical fiber cables. The upper assembly is based on the 60 mm Optical Cage System from Thorlabs. Currently, two mounting concepts are being considered for attaching this system to the Bosch aluminum profiles of the gondola.

In the first concept, the cage system is mounted on the inner side of the Bosch profiles, with the axicon extending beyond the profile boundary. The second concept positions the cage system on top of the Bosch profile. This configuration uses two mounting plates, one on the inner side and one on the outer side of the profile. While this option provides increased mechanical stability, it also results in additional weight.

The final selection between these two variants will be made based on appropriate simulations and experimental testing. If required, additional safety or support structures may be incorporated into the design.



Figure 4-1 Option 1: Internal mounting concept of the 60 mm cage system (side view)

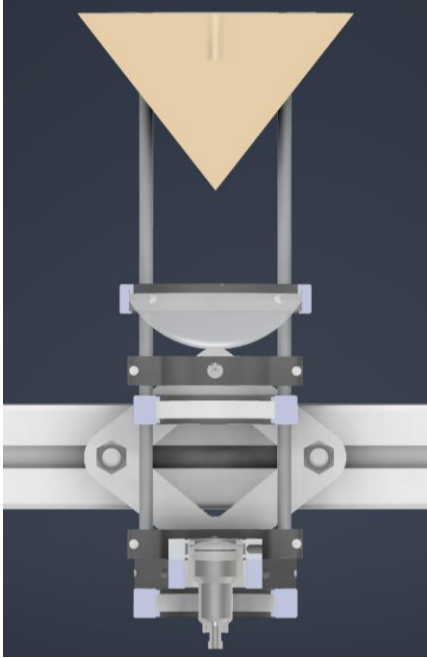


Figure 4-2 Option 1: Internal mounting concept of the 60 mm cage system (front view)



Figure 4-3 Option 2: Top-mounted cage system with dual mounting plates

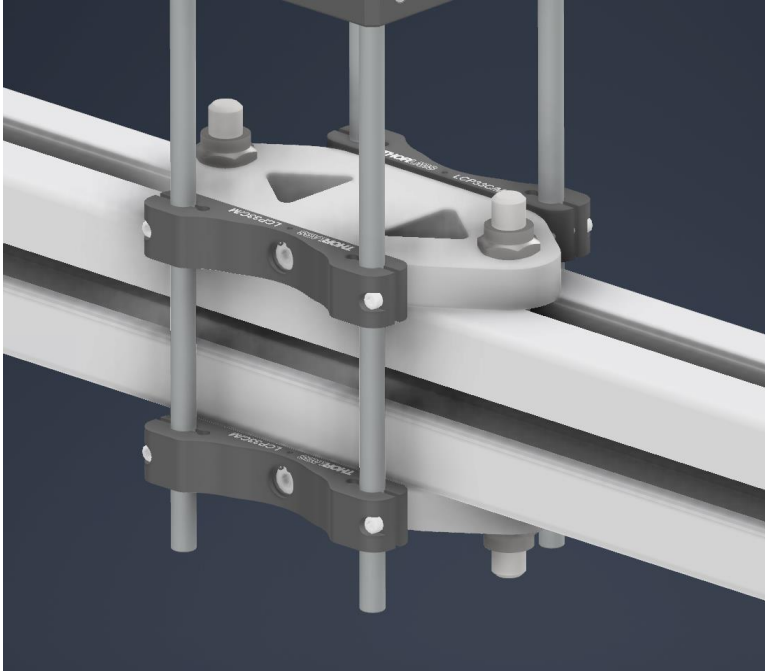


Figure 4-4 Option 2: Top-mounted cage system with dual mounting plates (detailed)

The lower assembly is mounted on two Bosch aluminum profiles that extend across the full width of the gondola. These profiles are mechanically connected to the gondola struts on both sides to ensure structural stability.

Aluminum mounting plates are installed on the Bosch profiles to serve as carrier structures for the remaining components. The measurement chamber and the electronics enclosure are positioned on these plates.

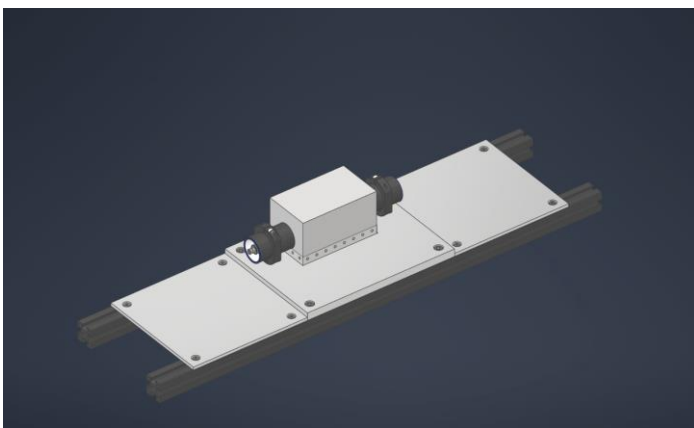


Figure 4-5 Lower Assembly Mounting on Bosch Profiles

4.2.2 Electrical

David

We do not require real-time data transmission, so the exact timing of when data is sent is not critical. We are therefore very flexible and can take other experiments into account. The downstream will use UDP, with a continuous data rate of about 2 kbit/s. After the start of our experiment, we may also occasionally send some larger packets. The upstream is minimal and consists only of control commands. For safety reasons, TCP will be used for this.

Our experiment will feature an Amphenol RJF21 connector with a CODE A insert as described in the BEXUS User Manual. For electrical power, our experiment will have one Amphenol PT02E8-4P connector with the assignment as specified in the User Manual (Pin A: +, Pin B: -, Pin C: empty, Pin D: empty). We will draw a maximum current of 1.2 A at 28 V and use a total energy of 121 Wh.

4.3 Experiment Components

Isabella

Table 4-1: Experiment summary table

Experiment mass (in kg):	10 kg
Experiment dimensions (for BEXUS) or module size (for REXUS) (in m):	Upper box: LWH 0,15 x 0,15 x 0,4 Lower box: LWH 0,6 x 0,25 x 0,3
Experiment expected COG (center of gravity) position:	Both parts have their center of gravity located approximately at their geometric center.

4.4 Optical Design

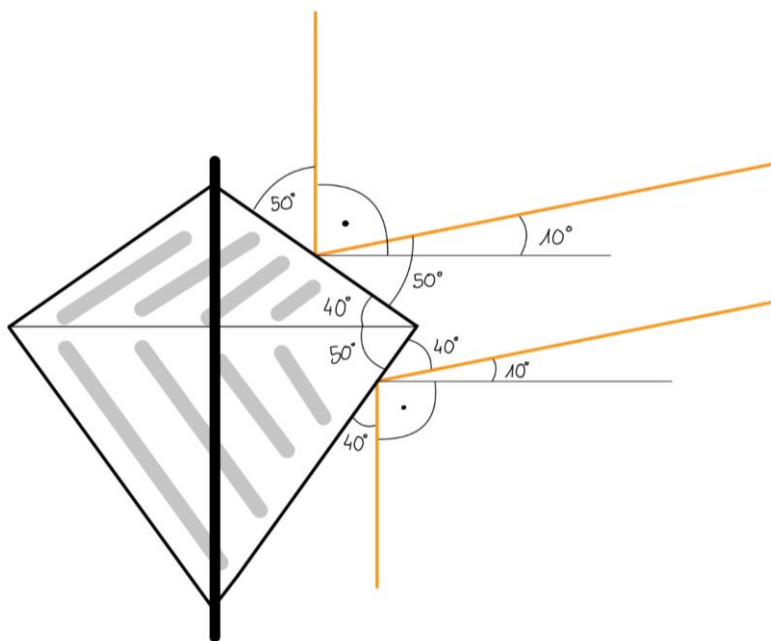
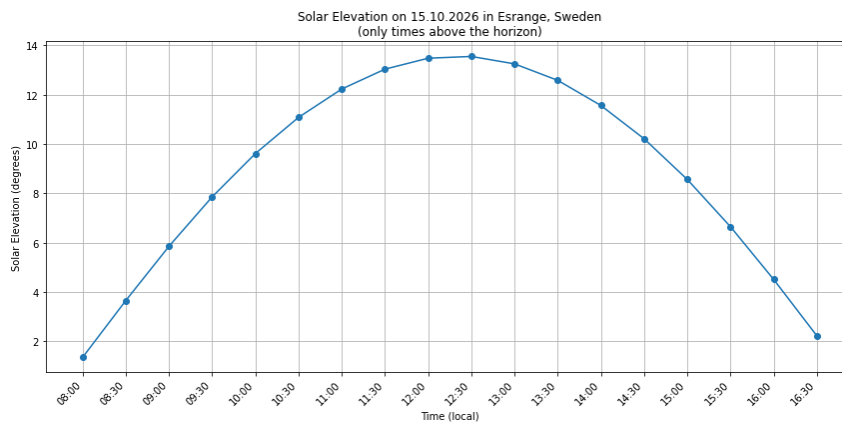
Isabella

REMARK: Consider doing a Ray Tracing Simulation (including the axicon). To get a better idea about the beam position and beam diameter in the chamber.

To obtain precise and meaningful optical measurements under continuously changing environmental conditions, including balloon rotation and motion, a parallel measurement approach is essential. For this purpose, the incoming sunlight is simultaneously divided into a measurement path and a reference path. While the measurement beam is guided through a chamber in which calcium carbonate can be released, the reference beam is directly coupled into the dual spectrometer.

The optical path is designed by following the sunlight from its source to the detection system. Due to the rotation of the balloon, the direction of incoming light changes continuously. To maintain a stable optical alignment independent of balloon orientation, a symmetric axicon is used to redirect the sunlight as vertically downward as possible. The axicon must be aligned according to the

solar position. Therefore, software-based solar position calculations are used, taking into account both the time of day and the altitude relative to ground level, to optimize the axicon orientation.



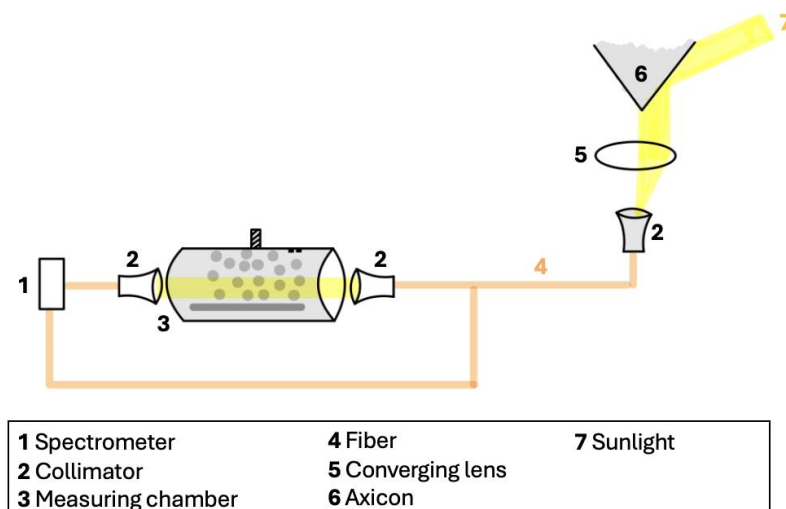
axis of symmetry (rotationally symmetric)

Although mechanical mounting of the axicon may introduce partial shading, such effects are minimized by using thin and symmetrically designed support

structures. Since both the reference and measurement paths originate from the same light source, partial shading is since it affects both paths the same.

The downward-directed beam is then focused by a converging lens and coupled into an optical fiber at the focal point using a collimator. A 1×2 multimode fiber optic coupler is used to split the optical signal into two fiber channels with a predefined intensity ratio. To enable reliable comparative measurements, the light intensity reaching the spectrometer from both fibers should be approximately balanced. A distribution of approximately 20% of the optical power to the reference path and 80% to the measurement path is considered a possible configuration. The final component selection will be determined after experimental testing.

The fiber of the reference path is directly connected to the spectrometer. In contrast, the measurement path is first expanded again using a collimator oriented in the opposite direction before entering the measurement chamber. The light passes through the chamber, where calcium carbonate can be released, and is subsequently recollected by a second collimator at the end of the chamber. Finally, the optical signal is guided via optical fiber into the dual spectrometer for analysis.



4.5 Mechanical Design

Kai

Mechanical – Detailed mechanical description, information about assembly and manufacturing, stress analysis and load calculations where applicable (Finite Element Analysis – FEA). CAD models and

datasheets must be included in Appendix C.

The mechanical design consists of two main assemblies. The upper assembly is dedicated to collecting sunlight. For the arrangement and alignment of the optical components, the Thorlabs 60 mm Optical Cage System is used.

At the top of the system, an axicon is installed. Due to its geometry, the axicon enables the collection of sunlight independently of the balloon's rotational orientation. Its precise shape is optimized according to the expected solar elevation angle, ensuring that the incoming rays are reflected as parallel as possible in the downward direction. The axicon will either be manufactured from aluminum or produced via 3D printing and coated with reflective mirror foil.

Below the axicon, a collection lens is mounted within the cage system using standard Thorlabs components. At a distance of exactly 8 cm, corresponding to the focal point of the collection lens, a collimator is positioned to couple the focused sunlight into an optical fiber. The cage rods are made of robust stainless steel, and the majority of the components are commercially available and mechanically compatible within the Thorlabs ecosystem. The custom-designed mounting interface to the Bosch aluminum profiles will most likely be manufactured from aluminum.



Figure 4-6 Upper Optical Collection Module with cage system

The optical fiber then splits the beam into two separate fibers using a 1×2 multimode fiber optic coupler. The optimal splitting ratio has yet to be determined. The objective is to achieve approximately equal light intensity in both fibers after calibration and during the measurement flight.

The reference path is directly coupled into a spectrometer. The measurement path, in contrast, is first diverged again before entering the measurement chamber. The light passes through the chamber, where calcium carbonate can be released, and is then recollected by a second collimator at the end of the chamber. Finally, the light is guided into a dual spectrometer.

The transition between the optical fibers and the chamber is realized using optical cubes. Inside these cubes, the collimators are securely mounted and protected by a permanently bonded glass window. Both optical cubes are designed to be easily removable as single units, allowing them to be inserted into or extracted from the chamber without complex disassembly.

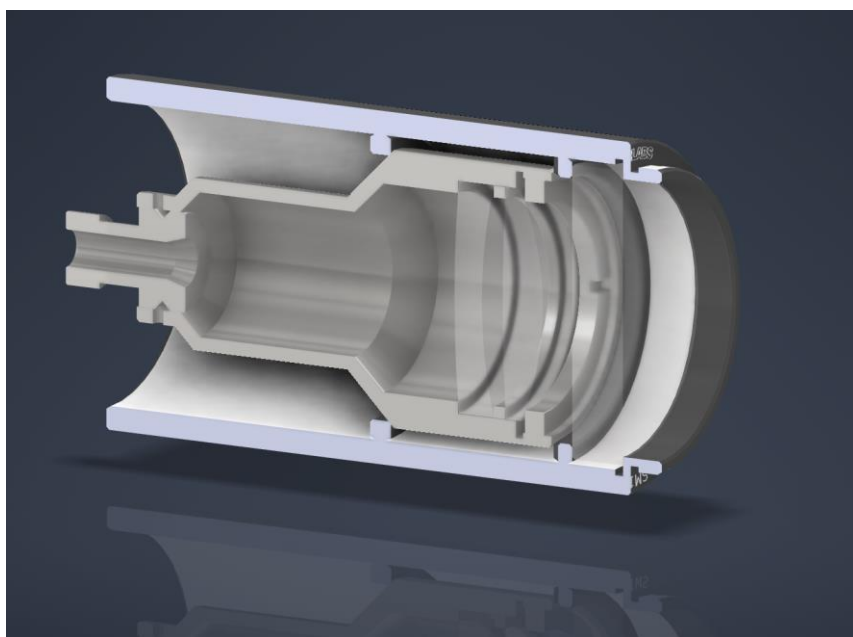


Figure 4-7 Optical Cube Interface Between Fiber and Measurement Chamber (cross-sectional view)

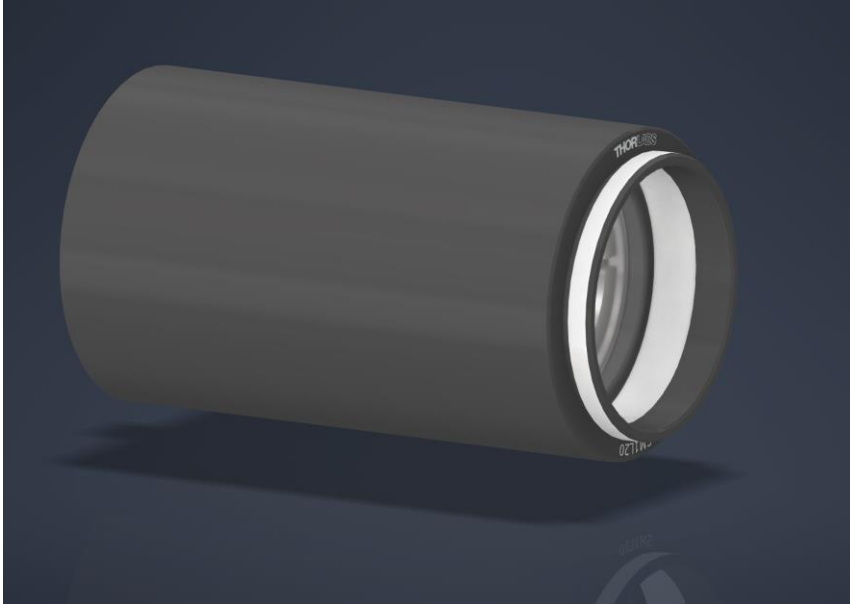


Figure 4-8 Optical Cube Interface Between Fiber and Measurement Chamber

Within the chamber itself, an additional protective glass window is bonded in place to shield the optical cubes from calcium carbonate contamination. The measurement chamber is open at the bottom and covered with a membrane that is tensioned along the outer walls. This membrane will most likely be made of silicone. The chamber is designed to be easily removable and will either be screwed or otherwise mechanically secured in place.

Below the test chamber, a push-pull solenoid is mounted. It strikes against the membrane to distribute the calcium carbonate evenly inside the chamber. At the top of the chamber, two pinch valves allow controlled release of calcium carbonate in two phases. Additionally, two solenoid valves are positioned at the side to establish defined atmospheric conditions within the chamber during ascent.

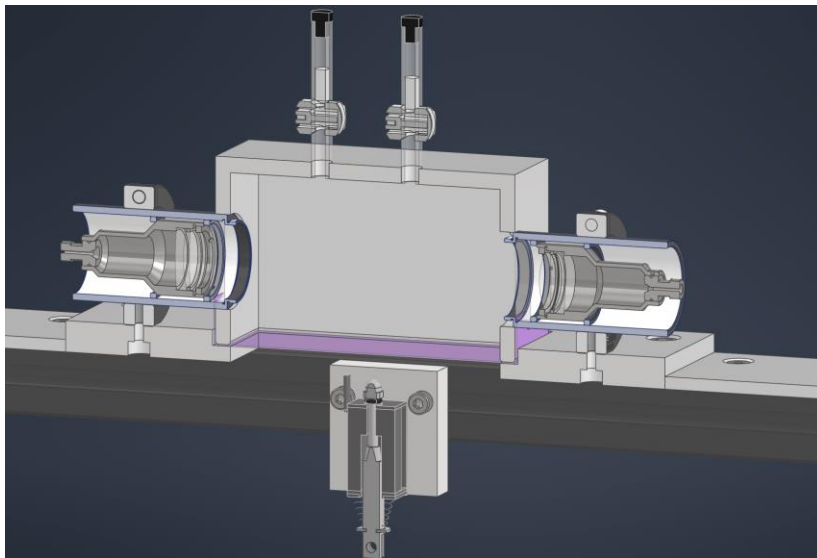


Figure 4-9 Measurement Chamber Assembly and Functional Components (cross-sectional view)

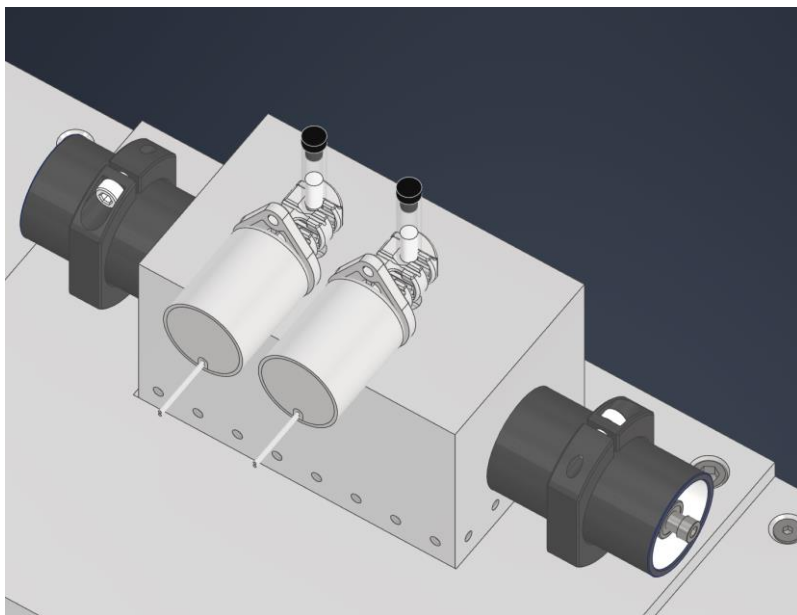


Figure 4-10 Measurement Chamber Assembly and Functional Components

Additionally, in order to avoid the need for installing valves directly within the chamber, the system is designed to use externally routed hoses with quick-connect fittings. The solenoid valves are therefore positioned at a slightly

greater distance from the chamber while maintaining controlled regulation of the internal atmospheric conditions during ascent.

The lower enclosure additionally contains the electronics compartment housing the Raspberry Pi. Environmental sensors (pressure, temperature, and particle counters) together with the inertial measurement unit (IMU) are strategically positioned within the structure to ensure representative and accurate measurement of the local environmental conditions.

The experimental setup is enclosed within a protective outer housing. Apart from meeting the structural requirements necessary to withstand balloon flight conditions, the enclosure is not subject to any additional functional constraints.

4.6 Electronics Design

4.7 Components

a) Sensors

Typ	Req.	Speq.	Operating conditions	Qty.
e9u-SPMD-350/850-10-Duo		350 – 850 nm 10nm 450 fps		1
STLM20W87F	-20 - +50°C +- 5°C	-20 - +100°C +-2.5°C	-55 - 130°C	2
BME280	0-100%	0 - 100% +- 1%	-40 - +85°C 300 – 1100hPa	1
Keller 23SY		0 – 1000 bar +- 0.4mbar	-40 - +85 °C	2

b) Actuators

Typ	Voltage	Power	Qty.
HS-1564B	12V	36W	1
USS-MSV00025	12V	2W	4

c) System

Typ	Function	Qty.
Raspberry pi 5	The Raspberry Pi 5 establishes the Ethernet connection and serves as the host for the spectrometer.	1

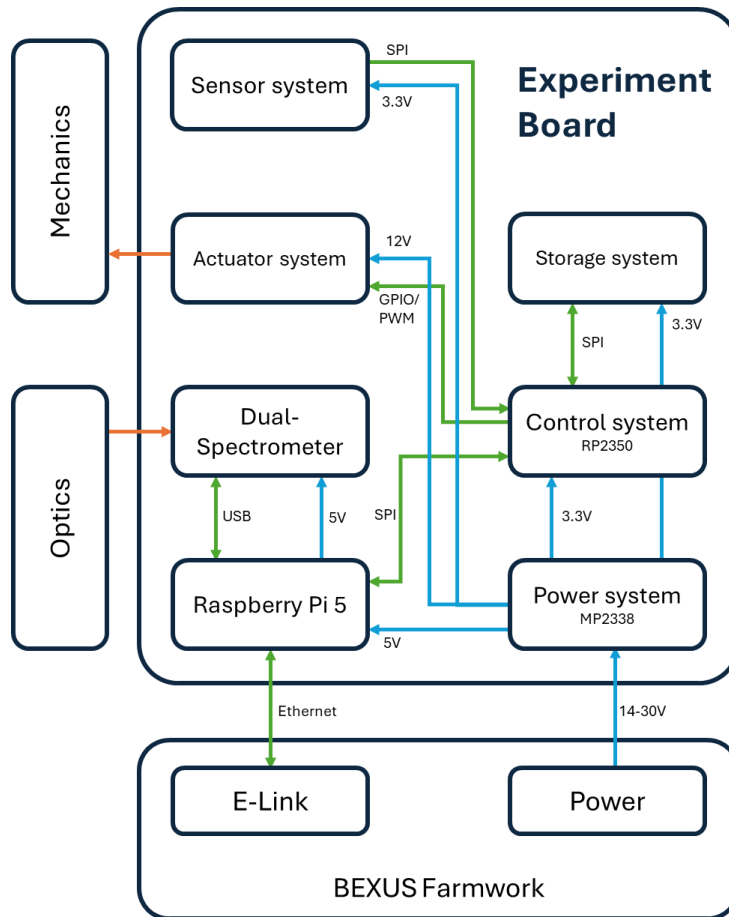
RP2350	The RP2350 handles sensor data acquisition as well as actuator control, ensuring that these functions operate independently of the Raspberry Pi's operating system	1
MP2338	The MP2338 converters provide the required voltage regulation for the three (or two) necessary bus voltages.	3
SD- Card	The collected data is stored redundantly on SD cards	3

4.8 System Architecture

The main electronics board is designed to accommodate a Raspberry Pi 5. The spectrometer is connected directly to the Raspberry Pi via USB, and the Ethernet interface is also connected directly to the Raspberry Pi.

All other subsystems are integrated on the main electronics board. These include the RP2350 microcontroller, which controls the experiment sequence and the actuators, the actuator driver circuitry, and the power subsystem that supplies the entire experiment with the required voltage rails.

The main electronics board is housed within a dedicated enclosure. Environmental sensors intended to measure external conditions are mounted on the outside of the enclosure. These sensors are either placed on a separate small sensor board or connected via short cables routed to the exterior of the housing.



a) Power

We will use the onboard power supply of the Gondola system via an Amphenol PT02E8-4P connector.

A 12 V power rail will be required for the actuators, while a 5 V bus will supply the Raspberry Pi 5 and the spectrometer.

The required voltage levels will be generated using switching regulators based on the MP2338.

A 3.3 V rail is required for the sensors and the RP2350. This rail will be implemented either with an LDO regulator or with an additional switching regulator, depending on efficiency and noise considerations.

Redundancy of the voltage rails via an emergency LDO (as discussed during the training week :) is planned.

Component	Voltage rail / V	Average operating current / A	Operating time	Energy / Wh
e9u-SPMD-350/850-10-Duo (Spectrometer)	5	0.25	3h	3.75
Sensors	3.3	0.1	5h	1.65
HS-1564B (Membrane)	12	2.5	2h	65
USS-MSV00025 (Valve)	12	0.17	30s (5s /operation)	0.02
Raspberry pi 5	5	0.8	5h	20
Miscellaneous			5h	5
Sum				96
Efficiency (85%)				111
Margin (10%)				121

b) Grounding

The experiment implements a Distributed Single Point Grounding (DSPG) strategy to ensure electrical stability, safety, and electromagnetic compatibility. A continuous PCB ground plane serves as the common reference potential for all analog, digital, and power circuits. The power return is connected to the PCB ground at a single defined point to control return current paths and prevent ground loops.

All metallic structural parts are mechanically connected to the gondola chassis, which provides the equipotential reference. No additional direct ground connections are implemented between individual metal parts and the PCB ground in order to avoid multiple return paths. This approach minimizes common-mode disturbances, prevents ground loops, and ensures safe dissipation of fault currents and electrostatic discharge.

c) Storage

The system includes three SD cards for data storage. One SD card is installed in the Raspberry Pi, while two additional SD cards are located on the main electronics board and are connected to the RP2350 via SPI. The two SD cards connected to the RP2350 are used for redundant data storage.

The Raspberry Pi stores the spectrometer measurement data and logs communication activities. The RP2350 stores the remaining

sensor data and logs actuator activity. This separation of storage responsibilities ensures structured data management and increases overall data reliability through redundancy.

d) Actuator

The experiment includes two types of actuators: electrically driven ball valves and a push-pull solenoid.

The ball valves are equipped with three wires: a common ground and two separate control lines for opening and closing. Each control line is driven by a MOSFET controlled via a GPIO pin of the RP2350. To prevent simultaneous activation of the open and close lines, hardware interlocking is implemented to ensure that both control signals cannot be driven high at the same time.

The push-pull solenoid is used to generate mechanical vibration and therefore requires a periodic control signal. The required 12 V periodic signal is generated using a switching stage with an inverter configuration and is controlled via a PWM signal from the RP2350. This allows precise control of the actuation frequency and duty cycle.

e) Microcontroller

tbd

f) Sensors

tbd

4.9 Thermal Design

Kai, David und Isabella

Thermal - a detailed multi-nodes thermal analysis is recommended, considering as possible nodes hot spots and components exposed to extreme hot/cold conditions. The use of a software package is recommended.

The CLOUDS experiment is designed to operate passively within the expected temperature ranges of the BEXUS flight profile, without any active thermal control (e.g., heaters or cooling systems). The total power dissipation is low (estimated <10 W average), and all major components (spectrometer, sensors, electronics, particle chamber) are selected with operating and survivable temperature ranges compatible with the anticipated stratospheric conditions (external temperatures typically between -40°C and +60°C, with internal temperatures moderated by self-heating and optional insulation).

The experiment will be enclosed in a simple insulated housing to reduce heat exchange with the cold ambient environment. During ascent, the open measurement chamber allows passive equilibration with external pressure and temperature. At float altitude, the chamber is sealed, and internal conditions are expected to remain within acceptable limits due to the low power consumption.

A basic thermal assessment (worst-case hot/cold scenarios) will be conducted prior to CDR to confirm adequate margins. The design ensures negligible thermal impact on the gondola or neighboring experiments, in compliance with BEXUS requirements.

This passive thermal design is similar to that used in previous low-power BEXUS experiments and will be verified through environmental testing (Section 5.2, T-04 Thermal Cycling).

4.10 Power System

The experiment is powered by the 28 V DC supply provided by the BEXUS gondola. The estimated total energy consumption of the experiment is approximately 121 Wh over the full duration of the flight.

Major power consumers include:

- Spectrometers and optical detectors
- Data acquisition and processing electronics
- Particle dispersion system (vibrating membrane)

The particle dispersion system may generate peak currents of up to approximately 1.2A, which are handled through appropriate electronic current limiting and control. No additional batteries or auxiliary power supplies are required.

The power budget includes sufficient margin to account for peak loads and ensures stable operation throughout the ascent, float, and descent phases of the flight.

4.11 Software Design

REMARK: Add more drawings, schematics, etc. (like flow charts in software section).

1. Purpose:

Experiment control (valve and membrane mechanism) and data handling of sensor data. Experiment sensor data shall be stored and transmitted to a ground station.

2. Design:

a) Process Overview

The experiment consists of a controlled release mechanism, a membrane ensuring homogeneous dispersion, optical sensors to capture measurement data, onboard data storage, and a downlink interface for transferring data to ground control.

The process begins with system initialization and self-diagnostics. Once triggered, potentially via remote activation, CaCO_3 is released and dispersed through the membrane. Optical sensors then measure optical properties (e.g. light intensity) over time. Sensor data is processed by the microcontroller, temporarily buffered, stored in non-volatile memory, and partially or fully transmitted to ground control via the downlink interface.

b) General and safety related concepts

Potential Error Behaviors:

- Sensor malfunction or invalid readings
- Communication failure between sensors and the microcontroller
- Storage writes errors or memory overflow
- Downlink transmission errors or data loss

Safety and Reliability Concepts:

- **Redundancy (Logical):** Critical sensor readings can be sanity-checked against predefined thresholds or expected ranges.
- **Fail-Safe States:** In case of critical failure, the system transitions to a safe state where actuators are disabled and data already collected is preserved.
- **Error Logging:** Errors and abnormal events are logged to non-volatile memory for post-mission analysis.
- **Watchdog Timer:** A hardware or software watchdog resets the microcontroller in case of software lock-up.
- **Checksum / CRC:** Data packets for storage and downlink include checksums to detect corruption.

c) Interfaces

Ethernet is used as the interface for external data communication.

For downstream data transmission, UDP is used as the transport protocol. Since occasional packet loss or minor data corruption does not critically affect the experiment results, UDP is preferred due to its lower overhead and reduced latency.

For upstream communication, TCP is used to ensure reliable and ordered data transmission. This guarantees that commands and critical configuration data are received completely and without corruption.

Different communication protocols are used within the electronics board. The Raspberry Pi is connected to the RP2350 microcontroller via UART for data exchange and control. SD cards are interfaced with the RP2350 using SPI. Environmental sensors are connected using their respective communication protocols. Actuators are controlled via PWM or GPIO signals as appropriate for each device.

The Raspberry Pi reads data from the spectrometer via USB, processes or formats the measurements, and transmits the results downstream over UDP. For upstream communication, the RP2350 periodically reads sensor values, stores or buffers them on the SD card if needed, and controls actuators according to the experiment sequence.

d) Data acquisition and storage

Since we don't yet have a final decision on the hardware, these estimations are difficult.

A rough estimation is given here:

- The optical sensors generate time-series measurement data during the experiment.
- Sampling rate: 100 samples/second
- Data size per sample: 16 bytes
- Maximum experiment duration: 300 minutes (meant to be too large)

Estimated Data Volume:

- $100 \text{ samples/s} \times 18000 \text{ s} \times 16 \text{ bytes} \approx 29,00,000 \text{ bytes} (\sim 29 \text{ MB})$

Storage Architecture:

- **RAM Buffer:** Temporary buffering of sensor data (e.g. circular buffer of 4–8 kB) to decouple acquisition from storage writes.
- **Permanent Storage:** Non-volatile memory (SD cards) used for long-term storage of raw (and potentially processed) data.
- **Redundant Storage:** All data is stored on both SD cards, which act as backup in case of malfunction of one.
- **Write Strategy:** Buffered block writes to minimize wear and reduce write latency. Data transmission rates to permanent storage are designed to exceed the sensor data rate, ensuring no buffer overflow under normal operating conditions.

e) Process Flow

The experiment follows a well-defined sequence of states:

Initialization

System boot

Interface and sensor initialization

Self-tests

Standby

Await start command from ground control or trigger event

Release Phase

Activate release mechanism

Verify actuation

Measurement Phase

Continuous sensor data acquisition

Buffering and storage

Optional downlink transmission

Termination

Stop data acquisition

Securely close storage files

Safe / Idle State

System monitoring

Await further commands

f) Modularization and pseudocode

Main Modules:

System Control Module: Handles state machine and overall coordination

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Sensor Module: Initializes sensors and retrieves measurement data

Actuator Module: Controls the release mechanism

Data Management Module: Buffering, storage, and data formatting

Communication Module: Handles downlink and command reception

Safety Module: Watchdog handling and error monitoring

Pseudocode:

MAIN()

 initialize_system()

 while (true)

 check_watchdog()

 update_state_machine()

STATE_MACHINE()

 switch (current_state)

 INIT:

 init_sensors()

 init_storage()

 current_state = STANDBY

 STANDBY:

 if start_command_received()

 current_state = RELEASE

 RELEASE:

 activate_release_mechanism()

 current_state = MEASUREMENT

 MEASUREMENT:

 data = read_optical_sensors()

 buffer_data(data)

 store_data_if_buffer_full()

 transmit_data_if_requested()

 if experiment_time_elapsed()

 current_state = TERMINATION

TERMINATION:

```
finalize_storage()  
current_state = SAFE
```

Implementation: We will be using C and C++ with PlatformIO and Pico SDK. Visual Studio Code will be our editor, we host the code on GitLab in a Git repository for collaboration.

4.12 Ground Support Equipment

The Ground Support Equipment (GSE) for the CLOUDS experiment is designed to facilitate pre-flight testing, debugging, calibration, and control of the experiment's subsystems. Given the experiment's relatively simple requirements—no fueling, gas handling, or complex mechanical loading is involved—the GSE is lightweight and portable, consisting primarily of a laptop-based system. This setup will be used during integration, ground testing, and late access at Esrange, allowing us to verify functionality, monitor data, and perform last-minute calibrations without requiring specialized infrastructure.

GSE Components

The GSE includes the following key components:

- **Laptop:** A standard team laptop (minimum specifications: 8 GB RAM, Intel Core i5 or equivalent, Windows/Linux OS) as the primary interface.
- **Interface Cables/Adapters:** Ethernet cables and adapters to connect to the experiment's central electronics unit (compatible with defined electrical interfaces in Section 4.5).
- **Power Supply Adapter:** A low-voltage DC power supply (e.g., 12–28 V, matching BEXUS gondola supply) to simulate flight power during bench tests.
- **Optional Peripherals:** External monitor for data visualization, multimeter for basic electrical checks, and a reference light source or sensor calibration tools (e.g., for temperature/humidity verification).

All components are team-provided, require no special handling/storage at Esrange, and comply with BEXUS power/safety limits.

4.12.1 GSE Software

The GSE software will consist of a custom graphical user interface (GUI) developed to support real-time data visualization, command execution, calibration, and logging.

We are currently evaluating two candidate technologies to determine the best fit for our needs:

- **LabVIEW:** LabVIEW enables rapid development of intuitive GUIs with native support for Ethernet communication, real-time data acquisition, and built-in tools for sensor displays, calibration interfaces, error logging, and command controls (e.g., manual triggering of particle release simulations). It has proven effective for similar GSE tasks in previous BEXUS experiments and the free LabVIEW Community Edition fully covers the required functionality for non-commercial use.
- **Python:** Python offers high flexibility through libraries such as PyVISA (instrument control), socket/Ethernet communication modules, Matplotlib/Plotly (real-time plotting), and PyQt/PySide (professional GUIs). It provides full open-source freedom, excellent version control compatibility, and is increasingly used in student projects.

Current Decision Status: The final choice between LabVIEW and Python will be made after PDR, based on team expertise, license availability, prototyping outcomes, and integration performance with the experiment's electronics. Initial prototyping will compare both options in terms of development speed, reliability, and ease of Ethernet communication with the central microcontroller/electronics unit.

Key software features (independent of the selected technology):

- Live display of sensor data (temperature, humidity, pressure, IMU, optical measurements).
- Manual control of test sequences (e.g., particle release simulation, chamber sealing).
- Calibration interfaces (e.g., offset adjustments, reference value comparisons).
- Error logging, command console, and data export (CSV/JSON format).
- Safety interlocks (e.g., prevention of ground-based particle release commands).

Development will proceed iteratively, beginning with basic monitoring functions in Q1 2026 and reaching full capability in time for ground verification tests (e.g., T-01 and T-03 in Section 5.2).

This GSE software approach will provide efficient, low-risk support for the CLOUDS experiment while remaining fully compliant with BEXUS interfaces and constraints. Further details, including the final technology selection, screenshots, and communication protocols, will be provided in subsequent SED versions.

4.12.2 GSE Operations and Interfaces

During ground operations, the GSE laptop will connect to the experiment via a dedicated external port on the electronics enclosure via Ethernet. This interface allows:

- Pre-integration functional testing and debugging in the university lab.
- On-site verification, calibration, and last-minute adjustments during late access at Esrange.
- Data download or diagnostics post-recovery if required.

The GSE operations are fully compatible with BEXUS requirements, including:

- Minimal power consumption (<50 W total for laptop and peripherals).
- No RF emissions or interference.
- Software-based interlocks to prevent unsafe commands (e.g., accidental particle release on ground).

All connections use standard, flight-approved cables and connectors. The GSE will be transported by the team and requires no special storage or handling at the launch site.

5 EXPERIMENT VERIFICATION AND TESTING

This chapter outlines the verification and testing approach for the CLOUDS experiment. At this early stage (SED v1.0), the design is preliminary and based on the experiment proposal. Verification activities will confirm that the experiment meets its scientific objectives, functional requirements, and BEXUS constraints (e.g., mechanical, electrical, thermal, and safety interfaces) as defined in Chapter 2.

The verification process follows a hierarchical approach: individual components are tested first, followed by subsystem integration, and finally full experiment-level testing. Tests will include functional checks, environmental simulations (e.g., thermal-vacuum, vibration), and performance validation under stratospheric-like conditions where feasible. All activities are planned to progressively close TBD items and reduce technical risk leading up to the Critical Design Review and flight campaign.

Verification methods include:

- **I** – Inspection
- **A** – Analysis
- **R** – Review of Design
- **T** – Test

Verification levels are defined as:

- **Component** – individual items
- **Subsystem** – integrated groups of components
- **Experiment** – full payload assembly

5.1 Verification Matrix

The verification matrix links each key requirement to a verification method (Inspection (I), Analysis (A), Review of Design (R), Test (T)) and the planned verification level (**Component**, **Subsystem**, **Experiment**). Success criteria are defined qualitatively at this stage; quantitative thresholds will be refined in later SED versions.

Table 5-1: Verification table

ID	Requirement text	Method	Reference	Status	Verification Result
F.1	The experiment shall measure the intensity and spectral distribution of solar radiation in the stratosphere.	T / A	T-01 (Optical/Spectral Calibration)	T – Pending A – Pending	Pending
F.2	The experiment shall disperse CaCO_3 particles uniformly within the measurement chamber to simulate stratospheric aerosol conditions.	T	T-02 (Particle Dispersion Test)	T – Pending	Pending
F.3	The experiment shall measure temperature and relative humidity during ascent and inside the measurement chamber.	T	T-03 (Environmental Sensor Calibration)	T – Pending	Pending
P.1	Intensity Measurement Accuracy [TBD]	T / A	T-01	T – Pending A – Pending	Pending

ID	Requirement text	Method	Reference	Status	Verification Result
P.2	Spectral Resolution / Discrete Spectral Sampling [TBD]	T	T-01	T – Pending	Pending
P.3	The spectrometer acquires data at a sampling rate of at least 1 Hz.	T	T-01	T – Pending	Pending
P.4	Spatial Uniformity of Particle Distribution [TBD]	T / A	T-02	T – Pending A – Pending	Pending
P.5	Temporal Uniformity of Particle Distribution [TBD]	T	T-02	T – Pending	Pending
P.6	The temperature measurement's range outside the balloon shall be between -20 °C and +50 °C.	I / T	T-03 / Datasheet	I – Pending T – Pending	Pending
P.7	The temperature measurements outside the balloon shall be made with an accuracy of ± 5 °C.	T	T-03	T – Pending	Pending

ID	Requirement text	Method	Reference	Status	Verification Result
P.8	The temperature measurement outside the balloon shall be made at a rate of 1 measurement every second (1Hz)	T	T-03	T – Pending	Pending
P.9	The humidity measurement range outside the balloon shall be between 0% and 100% relative humidity.	I / T	T-03 / Datasheet	I – Pending T – Pending	Pending
P.10	Accuracy Humidity Sensor [TBD, non-trivial for stratospheric conditions]	T	T-03	T – Pending	Pending
P.11	The humidity measurement outside the balloon shall be made at a rate of 1 measurement every second (1 Hz).	T	T-03	T – Pending	Pending
D.1	The experiment shall operate in the temperature profile of the BEXUS vehicle flight and launch.	T / A	T-04	T – Pending A – Pending	Pending

ID	Requirement text	Method	Reference	Status	Verification Result
D.2	The experiment shall operate in the vibration profile of the BEXUS vehicle flight	T	T-05	T – Pending	Pending
D.3	The experiment power connector shall be an Amphenol PT02E8-4P, Code A.	I	Inspection checklist	I – Pending	Pending
D.4	The experiment shall not disturb or harm the launch vehicle.	R / A	A-01 (FMECA)	R – Pending A – Pending	Pending
D.5	The experiment shall acquire data through parallel light measurements.	T	T-01	T – Pending	Pending
O.1	The experiment shall be designed for modular removal or expansion, allowing flexible integration and maintenance.	I / R	Design review	I – Pending R – Pending	Pending
O.2	The experiment shall be able to accept a request for radio silence at any time while on the launch pad.	T	T-07	T – Pending	Pending

ID	Requirement text	Method	Reference	Status	Verification Result
O.3	The experiment shall automatically release CaCO ₃ particles if the connection with the ground segment is lost or otherwise execute the full experiment sequence by default.	T	T-07	T – Pending	Pending
O.4	The experiment shall store all acquired data onboard the gondola and downlink a predetermined amount of data to the ground segment.	T	T-06	T – Pending	Pending
O.5	The experiment shall provide a simple manual mechanism for pressure equalization.	I / T	T-08	I – Pending T – Pending	Pending
C.1	The experiment shall not exceed a total mass of 10 kg.	I / A	T-09	I – Pending A – Pending	Pending
C.2	The experiment shall not consume more than 120 Wh of energy during the mission.	T / A	T-09	T – Pending A – Pending	Pending

5.2 Verification Plan

The verification plan outlines the specific tests and analyses intended to demonstrate compliance with the requirements listed in the Verification Matrix (Section 5.1). Each activity is referenced in the matrix and is designed to verify the experiment progressively at component, subsystem, and full experiment levels. The plan follows a proto-flight verification philosophy, where the flight model undergoes acceptance-level testing, with qualification-level margins applied where necessary.

All tests will be performed using appropriate laboratory equipment and environmental simulation facilities. Detailed test procedures, including setups, instrumentation, pass/fail criteria, data acquisition formats, and safety measures, will be developed before execution and documented in separate test reports. This plan is preliminary and will be updated in subsequent SED versions (post-PDR/CDR) with refined procedures, exact dates, and confirmed facilities.

Table 5-2: Test description for T-01 – Optical/Spectral Calibration

Test number	T-01
Test type	Functional / Calibration
Test facility	TBD
Tested item	Spectrometer, optical paths, reference light source
Model	Engineering Model / Breadboard
Procedure, test level, and duration	Calibrate intensity and spectral response against known source; verify parallel light coupling; acceptance level; duration TBD
Test campaign duration	TBD
Test campaign date	Q1–Q2 2026 (planned)
Test completed	No
Requirements verified	F.1, P.1–P.3, D.5

Table 5-3: Test description for T-02 – Particle Dispersion Test

Test number	T-02
Test type	Functional
Test facility	TBD
Tested item	Measurement chamber, particle dispenser, imaging system
Model	Engineering Model
Procedure, Test level, and duration	Release controlled amount of CaCO ₃ particles; assess spatial/temporal uniformity via imaging; acceptance level; duration TBD
Test campaign duration	TBD
Test campaign date	Q2 2026 (planned)
Test completed	No
Requirements verified	F.2, P.4–P.5

Table 5-4: Test description for T-03 – Environmental Sensor Calibration

Test number	T-03
Test type	Functional / Calibration
Test facility	TBD
Tested item	Temperature, humidity, pressure sensors
Model	Engineering Model
Procedure, test level, and duration	Expose sensors to controlled conditions; calibrate accuracy and sampling rates against references; acceptance level; duration TBD
Test campaign duration	TBD
Test campaign date	Q1–Q2 2026 (planned)
Test completed	No
Requirements verified	F.3, P.6–P.11

Table 5-5: Test description for T-04 – Thermal Cycling

Test number	T-04
Test type	Environmental
Test facility	TBD
Tested item	Full experiment assembly
Model	Flight Model
Procedure, test level, and duration	Cycle through the BEXUS temperature profile; monitor functionality and survival; qualification level; duration TBD
Test campaign duration	TBD
Test campaign date	Q2–Q3 2026 (planned)
Test completed	No
Requirements verified	D.1

Table 5-6: Test description for T-05 – Vibration Qualification

Test number	T-05
Test type	Environmental
Test facility	TBD
Tested item	Full experiment assembly
Model	Flight Model
Procedure, test level, and duration	Apply BEXUS vibration spectra (random/sine); inspect for damage; qualification level; duration TBD
Test campaign duration	TBD
Test campaign date	Q3 2026 (planned)
Test completed	No
Requirements verified	D.2

Table 5-7: Test description for T-06 – Electrical Interface & Power Test

Test number	T-06
Test type	Functional / Interface
Test facility	TBD
Tested item	Power connector, electronics, data storage/downlink
Model	Flight Model
Procedure, test level, and duration	Simulate BEXUS interfaces; measure power draw and data transfer; acceptance level; duration TBD
Test campaign duration	TBD
Test campaign date	Q3 2026 (planned)
Test completed	No
Requirements verified	D.3, O.4, C.2

Table 5-8: Test description for T-07 – Autonomy & Failsafe Test

Test number	T-07
Test type	Functional
Test facility	TBD
Tested item	Central control unit, particle release, full sequence
Model	Engineering Model
Procedure, Test level, and duration	Simulate comms loss; verify autonomous execution and radio silence; acceptance level; duration TBD
Test campaign duration	TBD
Test campaign date	Q2–Q3 2026 (planned)
Test completed	No
Requirements verified	O.2–O.3

Table 5-9: Test description for T-08 – Pressure Equalization Test

Test number	T-08
Test type	Functional
Test facility	TBD
Tested item	Chamber equalization mechanism
Model	Engineering Model
Procedure, Test level, and duration	Cycle pressure; verify manual sealing/opening without leaks; acceptance level; duration TBD
Test campaign duration	TBD
Test campaign date	Q2 2026 (planned)
Test completed	No
Requirements verified	O.5

Table 5-10: Test description for T-09 – Mass & Power Verification

Test number	T-09
Test type	Inspection / Analysis
Test facility	TBD
Tested item	Full experiment assembly
Model	Flight Model
Procedure, test level, and duration	Measure mass/COG; simulate power profile; acceptance level; duration TBD
Test campaign duration	TBD
Test campaign date	Ongoing/Q1–Q2 2026 (planned)
Test completed	No
Requirements verified	Inspection / Analysis

Table 5-11: Test description for T-10 – Full End-to-End Functional Test

Test number	T-10
Test type	Functional / Integration
Test facility	TBD
Tested item	Integrated experiment
Model	Flight Model
Procedure, test level, and duration	Simulate full mission sequence (ascent/float/release); verify end-to-end data flow; acceptance level; duration TBD
Test campaign duration	TBD
Test campaign date	Q3 2026 (planned)
Test completed	No
Requirements verified	All functional/operational requirements

Table 5-12: Test description for A-01 – Risk & Safety Analysis

Test number	A-01
Test type	Analysis
Test facility	N/A (document-based)
Tested item	Full experiment design
Model	Design documents
Procedure, test level, and duration	Perform FMECA; identify hazards and mitigations; qualitative review; duration TBD
Test campaign duration	TBD
Test campaign date	Q1 2026 (planned)
Test completed	No
Requirements verified	D.4

5.3 Verification Results

As of this SED version, no verification tests have been performed. All results are pending.

6 LAUNCH CAMPAIGN PREPARATION

6.1 Input for the Campaign / Flight Requirement Plans

6.1.1 Dimensions and Mass

Table 6-1: Experiment Mass and Volume

Experiment mass (in kg):	10
Experiment dimensions (for BEXUS) or module size (for REXUS) (in m):	Upper box: LWH 0,15 x 0,15 x 0,4 Lower box: LWH 0,6 x 0,25 x 0,3
Experiment expected COG (centre of gravity) position:	Both parts have their center of gravity located approximately at their geometric center.

6.1.2 Ground and Flight Safety Risks

Table 6-2: Ground and flight safety risks

Risk	Key Characteristics	Mitigation
Accidental release of CaCO_3	Calcium carbonate is a fine powder that could contaminate internal components or pose a minor inhalation risk if unintentionally released during ground handling or flight.	Only a small amount of CaCO_3 is used. The particles are fully contained within a dedicated sealed chamber.

6.1.3 Electrical Interfaces

Table 6-3: Electrical interfaces applicable to BEXUS

BEXUS Electrical Interfaces		
E-Link Interface: E-Link required: Yes		
	Number of E-Link interfaces:	1
	Number of required IP addresses:	2
	Data rate – downlink (max. and average):	400 Kbit/s Max and 100Kbit/s AVG
	Data rate – uplink (max. and average):	1 Kbit/s MAX
	Interface type (RS-232, Ethernet):	Ethernet
Power system: Gondola power required: Yes		
	Peak power and current consumption:	34W and 1.2A @ 28V
	Average power and current consumption:	24W and 0.86A @ 28V
	Total power and current consumption after lift-off	121Wh and 4.32Ah @ 28V
Power system: Experiment includes batteries: No		

6.1.4 Launch Site Requirements

Table 6-4: Launch Site Requirements

Category	Item	Amount
Workstation	Table	2
	Chair	4
	Whiteboard	1
	Power sockets	2
	LAN-port	1
Tools	Multimeter	1
	Laboratory power supply	1

6.1.5 Flight Requirements

Table 6-5: Flight Requirements applicable to BEXUS

Optimal altitude	20-30km
Preferred path	No preferred path
Minimum float time	1,5 h
Ground track length	No preferred ground track length
Light/dark condition	Light

6.1.6 Accommodation Requirements

For our experiment, it is optimal to utilize the full width of one of the gondola's narrow sides, with the axicon extending slightly beyond the gondola to properly focus the sunlight. The exact placement within this area is flexible and does not require the entire space currently illustrated in the schematic. In the layout, our experiment is represented as the larger enclosure on the right, with the axicon



6.2 Preparation and Test Activities at Esrange

zusammen sammeln Elektroniktest: Laptop (Netzteil oder Versorgungsmodul)
10 Minuten (Spektrometer, alle Sensoren (Kommunikation))

Ventiltest+ Membrantest

(wenn müssen Kalkfreilassung) was zum Putzen?

Lichtpfad test

Is there any remove before flight part? No

- Is there any special recovery operation to be performed by the recovery team?No
- Is there any special handling procedure during motor mating, roll out, loading and/or countdown?No
- Does the team need a late access? Eventuell
- Is there a temperature range to be respected during operations?No

6.3 Timeline for Countdown and Flight

zusammen sammeln

6.4 Post-Flight Activities

zusammen sammeln

7 DATA ANALYSIS AND RESULTS

7.1 Data Analysis Plan

Mathis

Overview:

1. Data integrity, synchronization, and calibration checks
2. Exploratory visualization (time-series, differences, distributions)
3. Missing data assessment and justified handling
4. Independence assessment
5. Definition and verification of statistical assumptions
6. Primary analysis: paired comparison of control vs. test
7. Sensitivity analysis (non-parametric / robust methods if needed)
8. Effect size estimation and uncertainty
9. Power and limitation discussion
10. Statistical review from advisor

Explanation:

Before analysis, time synchronization between the two chambers is verified, and sensor calibration consistency is checked to account for possible offsets or drift.

An initial exploratory data analysis is performed to assess the recorded data visually. This includes time-series plots of the control and test chamber measurements, direct comparison plots, and plots of the difference (test minus control). These visualizations are used to identify trends, transient effects, and potential anomalies.

The extent and pattern of missing data are then assessed to determine whether missingness is random or related to known experimental or sensor-related events. Missing values are handled in a manner consistent with the physical interpretation of the measurements, with exclusion or interpolation applied only when justified and documented.

Because the measurements form a time series, the independence of observations is explicitly evaluated. Autocorrelation and time-series plots are used to assess dependence within each chamber's data. If significant dependence is detected, analysis methods that account for temporal dependence will be used.

The distributional properties of the data are then assessed. Normality is evaluated primarily for the paired difference between test and control measurements, using visual tools such as histograms and Q-Q plots, supplemented by formal tests where appropriate. These assessments determine the suitability of parametric statistical tests.

Potential outliers are identified using predefined criteria, such as known sensor saturation limits or documented experimental disturbances. Outlier handling procedures are established before hypothesis testing, and sensitivity analyses are conducted where appropriate to assess the impact of outliers on the results.

The primary statistical analysis compares the control and test chamber measurements using paired statistical tests applied to synchronized data points. A paired Student's t-test is used to assess whether the mean difference between test and control measurements is significantly different from zero, provided that the assumptions of normality and independence are satisfied. If these assumptions are violated, a non-parametric alternative, such as the Wilcoxon signed-rank test, is employed.

The significance level (α) is defined before analysis, typically at $\alpha = 0.05$. Effect sizes and confidence intervals are calculated alongside p-values to quantify the magnitude and uncertainty of observed differences. Power considerations are discussed in terms of the detectable effect size and the limitations imposed by the sample size.

Finally, the analysis procedures, assumptions, and results are reviewed by a statistics advisor at our university before and after to ensure methodological rigor and transparency.

7.2 Launch Campaign

To be completed in future SED versions.

7.3 Results

To be completed in future SED versions.

7.4 Input to Campaign Report

To be completed in future SED versions.

Lessons Learned Alle Sammeln Mathis

- Documentation is a difficult and time-consuming part but helps with communication inside and outside the team
- Difficulty of having an exact mass calculation
- Asking for help on time is essential and solves many problems before they occur
- Reading requirements in detail is not only good but essential and can't be skipped or done later

8 ABBREVIATIONS AND REFERENCES

8.1 Abbreviations

AIT	Assembly, Integration and Test
ASAP	as soon as possible
CaCO ₃	Calcium Carbonate
CDR	Critical Design Review
COG	Centre of Gravity
CRP	Campaign Requirement Plan
DLR	Deutsches Zentrum für Luft- und Raumfahrt
EAT	Experiment Acceptance Test
EAR	Experiment Acceptance Review
ECTS	European Credit Transfer System
EIT	Electrical Interface Test
EPM	Espace Project Manager
ESA	European Space Agency
Espace	Espace Space Center
ESTEC	European Space Research and Technology Centre, ESA (NL)
ESW	Experiment Selection Workshop
FAR	Flight Acceptance Review
FST	Flight Simulation Test
FRP	Flight Requirement Plan
FRR	Flight Readiness Review
GSE	Ground Support Equipment
HK	House Keeping
H/W	Hardware
ICD	Interface Control Document
I/F	Interface
IPR	Integration Progress Review
LO	Lift Off
LT	Local Time
LOS	Line of Sight
Mbps	Mega Bits per second
MFH	Mission Flight Handbook
MORABA	Mobile Raketen Basis (DLR, EuroLaunch)
OP	Oberpfaffenhofen, DLR Center
PCB	Printed Circuit Board (electronic card)
PDR	Preliminary Design Review
PST	Payload System Test
RBF	Remove Before Flight
SED	Student Experiment Documentation
SRM	Solar Radiation Management
SNSA	Swedish National Space Agency

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SODS	Start Of Data Storage
SOE	Start Of Experiment
SSC	Swedish Space Corporation
STW	Student Training Week
S/W	Software
T	Time before and after launch noted with + or -
TBC	To be confirmed
TBD	To be determined
WBS	Work Breakdown Structure
ZARM	Zentrum für Angewandte Raumfahrttechnologie und Mikrogravitation

8.2 References

APPENDIX A - EXPERIMENT REVIEWS

Preliminary Design Review – PDR

Critical Design Review – CDR

Integration Progress Review – IPR

Experiment Acceptance Review – EAR

APPENDIX B – OUTREACH AND MEDIA COVERAGE

APPENDIX C – ADDITIONAL TECHNICAL INFORMATION

APPENDIX D – CHECKLISTS